

UNIVERSITI TEKNOLOGI MARA

FACULTY OF BUSINESS AND MANAGEMENT

FIN659

QUANTITATIVE RESEARCH METHODS

GROUP ASSIGNMENT - DATA ANALYSIS

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BA2805C (GROUP 7)

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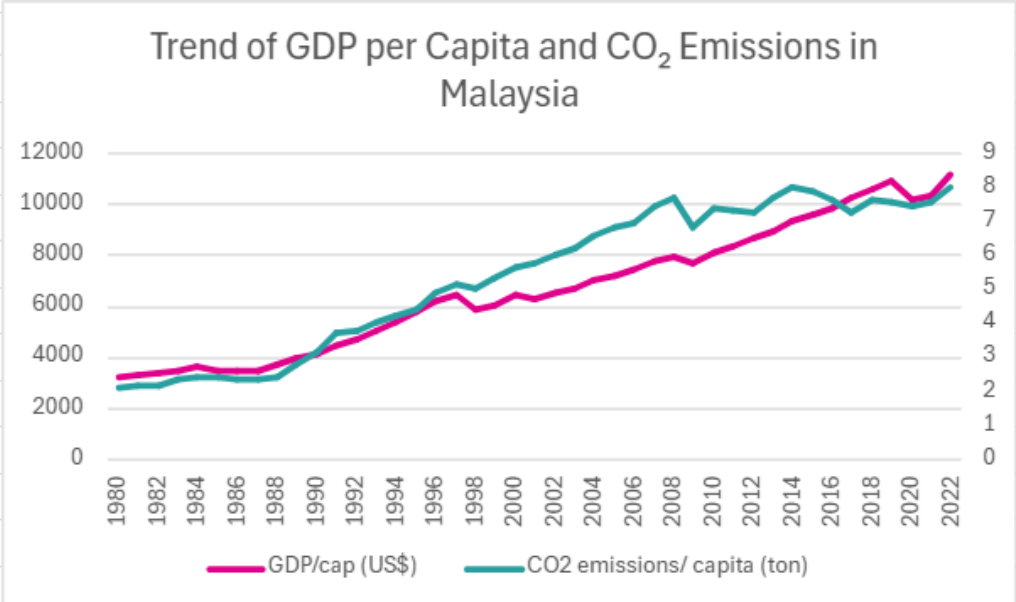
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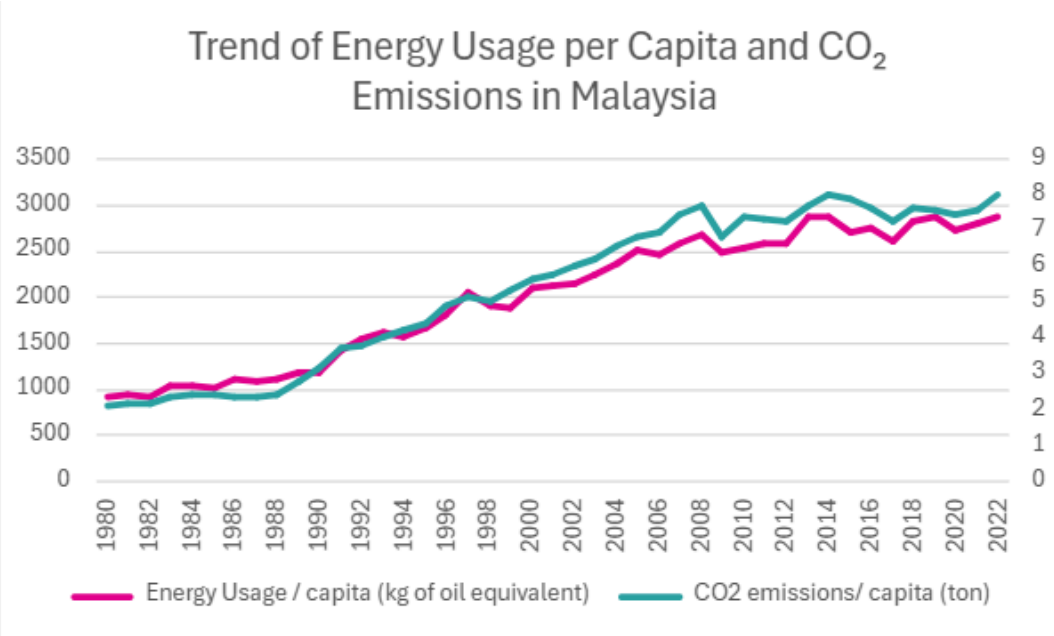
1.0 JUSTIFICATION OF VARIABLES

1.1 Gross Domestic Product per Capita (GDP/cap)



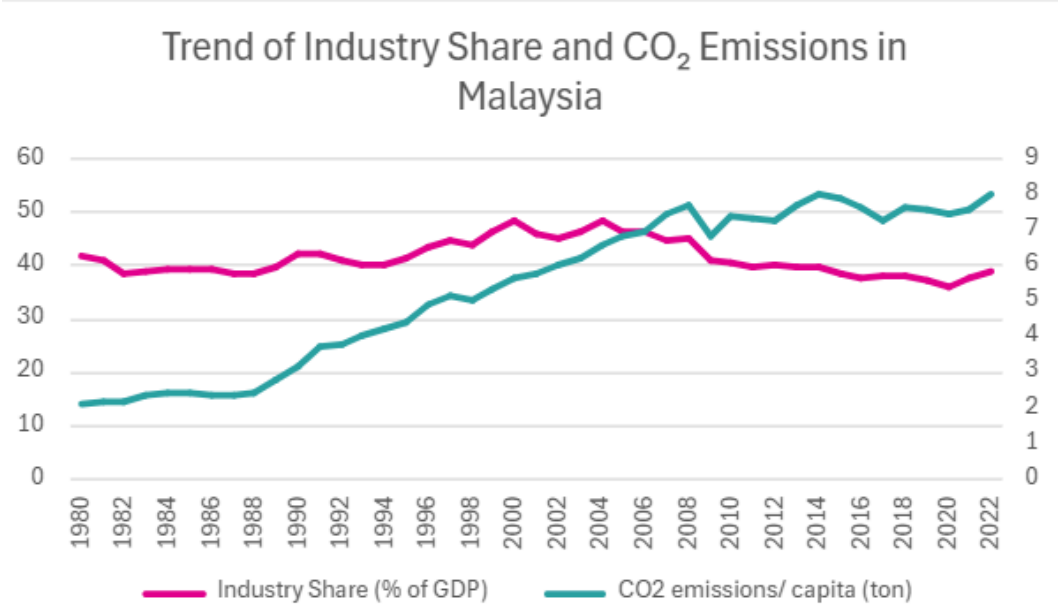
GDP per capita is a key indicator of Malaysia's economic progress and the rising standard of living for its citizens. When we look at the trend from 1980 to 2022, there is a clear correlation where carbon emissions increase alongside economic wealth. This happens because a growing economy requires a massive expansion in infrastructure, industrial production, and energy consumption to sustain its momentum. As personal incomes rise, there is a significant shift in lifestyle, households consume more electricity and the demand for private vehicle ownership increases. In Malaysia's case, because much of the energy needed to power this growth has historically come from fossil fuels, the rise in GDP has directly resulted in a higher carbon footprint. Essentially, the scale of the economy has expanded so much that the environmental cost, in the form of CO_2 emissions, has become a natural byproduct of the country's development.

1.2 Energy Usage per Capita (kg of oil equivalent)



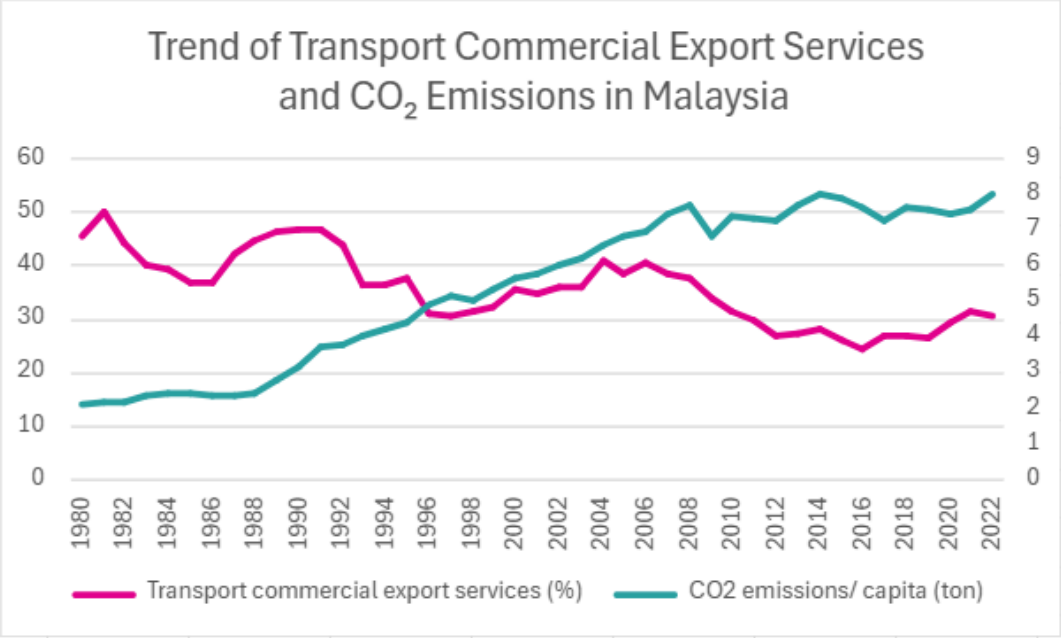
Energy usage per capita is perhaps the most direct driver of carbon emissions in Malaysia. When reviewing the data from 1980 to 2022, there is a very tight relationship where carbon output rises almost exactly in line with the amount of energy being consumed. This is largely because Malaysia's energy grid and industrial sectors still rely heavily on the combustion of fossil fuels, such as coal and natural gas. As the population uses more energy for daily activities ranging from cooling homes with air conditioning to powering large-scale manufacturing, more of these fuels must be burned, which releases CO_2 into the atmosphere as a byproduct. The trend in the chart confirms that as long as energy needs continue to grow to support a modern lifestyle, the environmental pressure from emissions will likely continue to climb as well.

1.3 Industry Share (% of GDP)



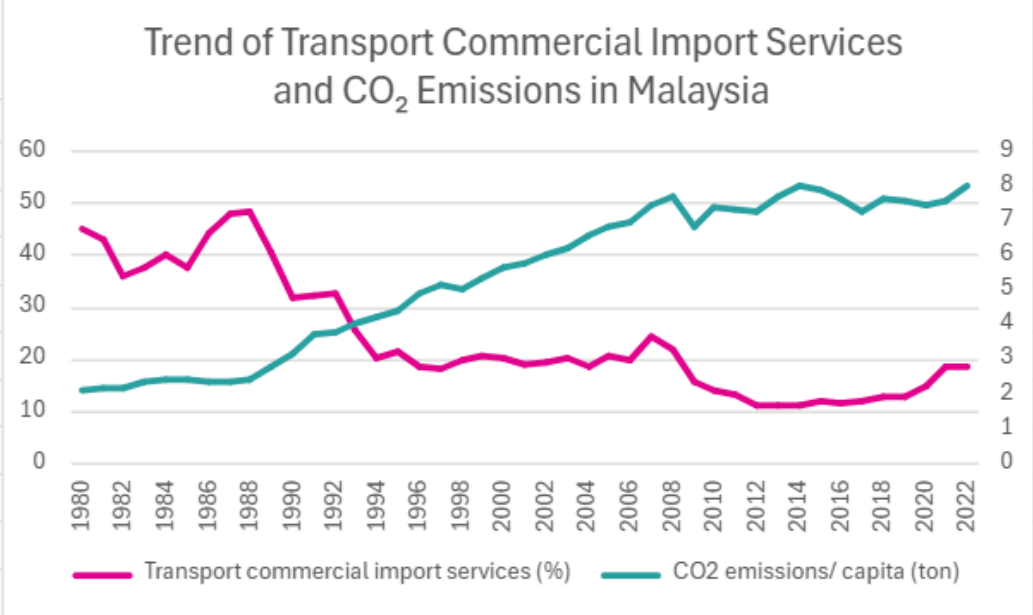
The industrial sector, which includes manufacturing, mining, and construction, is a core component of Malaysia's economic structure. This sector is highly energy-intensive, often requiring heavy machinery and high-heat processes that consume large amounts of fuel and electricity. While the industry share as a percentage of GDP has seen fluctuations between roughly 35% and 48% over the years, it remains a primary source of carbon output. Even when the sector's share of GDP remains steady or slightly declines, the total volume of industrial activity usually grows alongside the national economy. This sustained activity leads to continuous emissions from factory operations and industrial waste. Therefore, the industrial sector is a significant factor in Malaysia’s carbon footprint, as the nature of these activities inherently involves a higher degree of pollution compared to the service or agricultural sectors.

1.4 Transport Commercial Export Services (%)



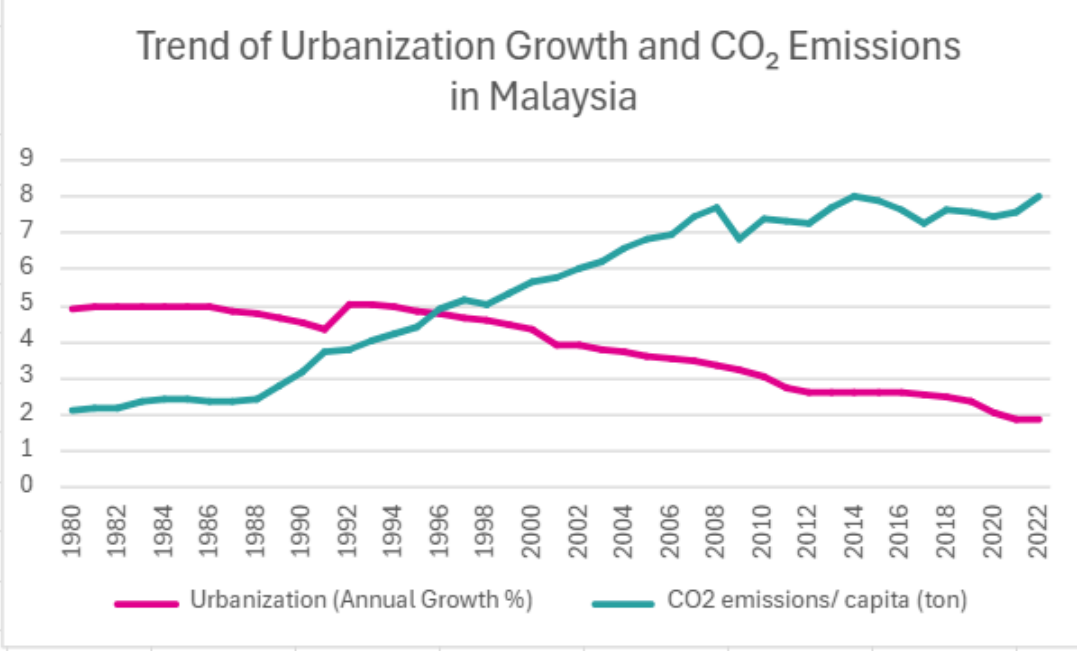
Transport commercial export services represent the logistics and shipping activities required to move Malaysian goods to international markets. Even though the percentage share of these services has shifted over the years, the total volume of trade has grown, requiring massive fuel consumption from cargo ships, heavy-duty trucks, and air freight. These transport modes are heavily dependent on fossil fuels, such as diesel and jet fuel, which release significant amounts of CO_2 during combustion. Therefore, the infrastructure and energy needed to support Malaysia's global export reach directly contribute to the national carbon footprint, as higher trade activity generally demands more energy-intensive logistics.

1.5 Transport Commercial Import Services (%)



Transport commercial import services are essential for bringing foreign goods, raw materials, and machinery into Malaysia to support domestic consumption and industrial needs. Similar to export logistics, the movement of these goods through international shipping, air freight, and trucking relies heavily on the combustion of carbon-intensive fossil fuels. Even though the percentage share of these services relative to other sectors may fluctuate or decline as seen in the chart, the total physical volume of imports often grows as the economy expands. This increased activity necessitates higher energy consumption for transport and warehousing, which directly translates to a rise in total carbon emissions. Therefore, the logistical demand created by importing goods is a significant contributor to Malaysia's environmental pressure and overall carbon footprint.

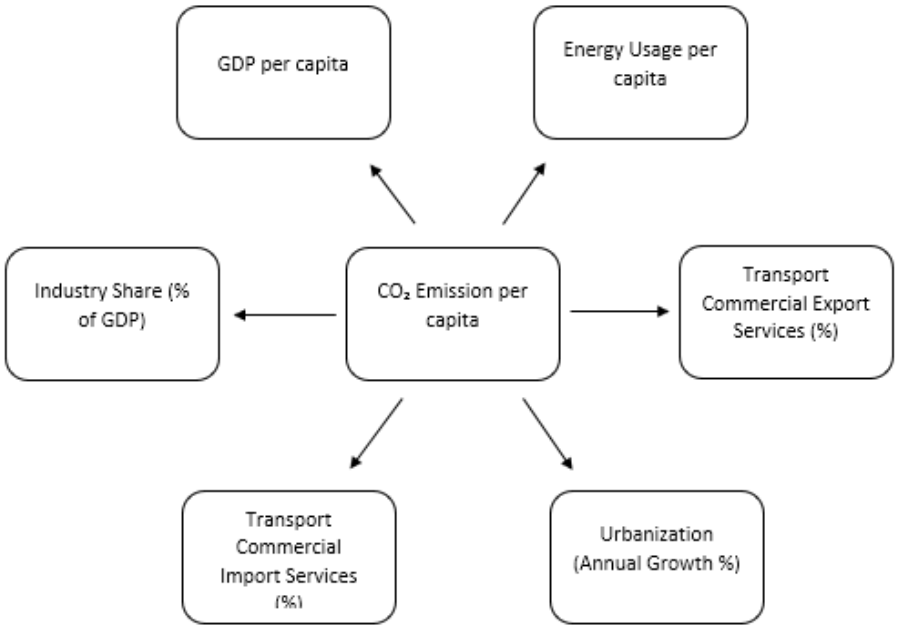
1.6 Urbanization (Annual Growth %)



Urbanization represents the physical growth of urban areas as the population moves from rural to city centers. Although the annual growth rate of urbanization in Malaysia has gradually declined over the decades as shown in the chart, the total number of people living in cities continues to rise. Urban living typically leads to higher carbon emissions because city residents have higher energy demands for air conditioning, lighting, and appliances. Furthermore, urbanization drives the expansion of infrastructure, such as the construction of high-rise buildings and transit networks, which are energy-intensive processes. The increase in urban density also leads to higher concentrations of private vehicle usage. Consequently, as Malaysia becomes more urbanized, the concentrated demand for energy and transport resources leads to a persistent increase in CO_2 emissions per capita, making it a key factor in environmental degradation.

2.0 THEORETICAL FRAMEWORK

2.1 Conceptual Diagram



2.2 Explanation of Relationships

Explain how each variable affects CO_2

- **GDP per capita:**

  CO_2 emissions tend to increase as GDP per capita rises over time. As countries develop economically, they increase and improve their overall output of energy, leading to greater use of carbon-based fuels (Grossman & Krueger, 1995). During the early and middle stages of economic development, rising GDP per capita is therefore associated with higher carbon emission levels.

- **Energy usage per capita:**

  Higher per capita energy consumption, particularly from fossil fuel sources contributes significantly to increased CO_2 emissions. Per capita energy use is known widely as one of the key determinants of CO_2 emissions over time, especially in developing and emerging economies (Stern, 2004).

- **Industry share (% of GDP):**

  An increasing share of the industrial sector in GDP represents an increase in the amount of manufacturing and processing activity, as this type of activity is generally energy-intensive. The growth of the industry will lead to an increase in CO_2 emissions through the combustion of increasing amounts of fuel and manufacturing output (Zhang & Cheng 2009).

- **Transport commercial export services (%):**

  The increase in transportation services for exports leads to higher levels of logistics activities, fuel consumption and freight movement. These increases will have a direct and positive relationship with CO_2 emissions (Saidi & Hammami 2015).

- **Transport commercial import services (%):**

  Similar to transportation services for exports, an increase in transportation services for imports will involve all forms of transportation including maritime, air, and road transports, all of which rely heavily on fossil fuels, thus increasing CO_2 emissions as a result of increased transportation services for imports (Baek et al., 2009).

- **Urbanization (annual growth %):**

  Due to higher electricity consumption and infrastructure development, urbanization is a direct contributor to increased population density and demand for transportation. Therefore, rapid urban growth will lead to higher CO_2 emissions through increased energy consumption in urban areas (Poumanyvong & Kaneko 2010).

3.0 REGRESSION MODEL

3.1 Model Equation

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Urbanization Growth (%), Industry share (%), Transport export services (%), Transport import services (%), Energy usage per capita, GDP per capita[b]	.	Enter

a. Dependent Variable: CO2 Emission per capita

b. All requested variables entered.

Model Summary[b]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Durbin-Watson
1	.997[a]	.994	.993	.18026	1.688

a. Predictors: (Constant), Urbanization Growth (%), Industry share (%), Transport export services (%), Transport import services (%), Energy usage per capita, GDP per capita

b. Dependent Variable: CO2 Emission per capita

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	189.515	6	31.586	972.029	.000[b]
	Residual	1.170	36	.032		
	Total	190.685	42			

a. Dependent Variable: CO2 Emission per capita

b. Predictors: (Constant), Urbanization Growth (%), Industry share (%), Transport export services (%), Transport import services (%), Energy usage per capita, GDP per capita

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients			Collinearity Statistics	
		B	Std. Error	Beta	t	Sig.	Tolerance	VIF
1	(Constant)	.291	.957		.304	.763		
	GDP per capita	-5.753E-5	.000	-.067	-.825	.415	.026	38.753
	Energy usage per capita	.003	.000	.885	11.064	.000	.027	37.507
	Industry share (%)	.025	.016	.039	1.601	.118	.290	3.451
	Transport export services (%)	.009	.009	.029	.985	.331	.202	4.959
	Transport import services (%)	-.019	.007	-.101	-2.599	.013	.112	8.919
	Urbanization Growth (%)	-.231	.108	-.115	-2.128	.040	.058	17.165

a. Dependent Variable: CO2 Emission per capita

Regression equation:

CO_2 emissions per capita = 0.2910 - 0.00005753 (GDP) + 0.003 (Energy) + 0.025(Industry share) + 0.09(Transport export services) - 0.019 (Transport import services) - 0.231 (Urbanization growth)

3.2 t-Statistics and p-Values

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients			Collinearity Statistics	
		B	Std. Error	Beta	t	Sig.	Tolerance	VIF
1	(Constant)	.291	.957		.304	.763		
	GDP per capita	-5.753E-5	.000	-.067	-.825	.415	.026	38.753
	Energy usage per capita	.003	.000	.885	11.064	.000	.027	37.507
	Industry share (%)	.025	.016	.039	1.601	.118	.290	3.451
	Transport export services (%)	.009	.009	.029	.985	.331	.202	4.959
	Transport import services (%)	-.019	.007	-.101	-2.599	.013	.112	8.919
	Urbanization Growth (%)	-.231	.108	-.115	-2.128	.040	.058	17.165

a. Dependent Variable: CO2 Emission per capita

The t-statistics and p-values are used to test the significance of each independent variable. A variable is considered significant if the p-value is less than 0.05.

From the regression results, energy usage per capita is statistically significant (t = 11.064, p = 0.000). This indicates that energy usage per capita has a significant effect on CO_2 emissions

per capita. In addition, transport import services ($t = -2.599$, $p = 0.013$) and urbanization growth ($t = -2.128$, $p = 0.040$) are also statistically significant, showing that both variables significantly influence CO_2 emissions per capita.

However, p-values greater than 0.05 indicate that GDP per capita ($p = 0.415$), industry share ($p = 0.118$), and transport export services ($p = 0.331$) are not statistically significant.

3.3 Null and Alternate Hypotheses for Each Variable

The alternative hypothesis (H1) asserts that the independent variable has an impact on CO_2 emissions per capita, whereas the null hypothesis (H0) claims that it has no effect.

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients			Collinearity Statistics	
		B	Std. Error	Beta	t	Sig.	Tolerance	VIF
1	(Constant)	.291	.957		.304	.763		
	GDP per capita	-5.753E-5	.000	-.067	-.825	.415	.026	38.753
	Energy usage per capita	.003	.000	.885	11.064	.000	.027	37.507
	Industry share (%)	.025	.016	.039	1.601	.118	.290	3.451
	Transport export services (%)	.009	.009	.029	.985	.331	.202	4.959
	Transport import services (%)	-.019	.007	-.101	-2.599	.013	.112	8.919
	Urbanization Growth (%)	-.231	.108	-.115	-2.128	.040	.058	17.165

a. Dependent Variable: CO2 Emission per capita

- **GDP per capita**

  H0: GDP per capita has no significant effect on CO_2 emissions per capita.

  H1: GDP per capita has a significant effect on CO_2 emissions per capita

  Decision: Accept H0 (p-value: 0.415 > 0.05)

  Conclusion: GDP per capita has no significant effect on CO_2 emissions per capita.

- **Energy usage per capita**

  H0: Energy usage per capita has no significant effect on CO_2 emissions per capita.

  H1: Energy usage per capita has a significant effect on CO_2 emissions per capita.

  Decision: Reject H0, (p-value: 0.000 < 0.05)

  Conclusion: Energy usage per capita has a significant effect on CO_2 emissions per capita.

- **Industry share (% of GDP)**

  H0: Industry share has no significant effect on CO_2 emissions per capita.

  H1: Industry share has a significant effect on CO_2 emissions per capita.

  Decision: Accept H0,(p-value = 0.118 > 0.05)

  Conclusion: Industry share has no significant effect on CO_2 emissions per capita.

- **Transport export services**

  H0: Transport export services have no significant effect on CO_2 emissions per capita.

  H1: Transport export services have a significant effect on CO_2 emissions per capita.

  Decision: Accept H0, (p-value = 0.331 > 0.05)

  Conclusion: Transport export services has no significant effect on CO_2 emissions per capita.

- **Transport import services**

  H0: Transport import services have no significant effect on CO_2 emissions per capita.

  H1: Transport import services have a significant effect on CO_2 emissions per capita.

  Decision: Reject H0, (p-value = 0.013 < 0.05)

  Conclusion: Transport import services has a significant effect on CO_2 emissions per capita.

- **Urbanization growth**

  H0: Urbanization growth has no significant effect on CO_2 emissions per capita.

  H1: Urbanization growth has a significant effect on CO_2 emissions per capita.

  Decision: Reject H0, (p-value: 0.040 < 0.05)

  Conclusion: Urbanization growth has a significant effect on CO_2 emissions per capita.

4.0 DIAGNOSTIC CHECKS FOR TIME SERIES DATA

4.1 Autocorrelation

Model Summary[b]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Durbin-Watson
1	.997[a]	.994	.993	.18026	1.688

a. Predictors: (Constant), Industry, Energy, Export, Import, Urbanization, GDP

b. Dependent Variable: CO2

Based on our SPSS regression results, the Durbin-Watson statistic is 1.688. Since this value is close to the threshold of 2.0, we can conclude that there is no significant autocorrelation present in the model's residuals. This indicates that the error terms for each year are independent of one another, satisfying a major assumption for linear regression and confirming that our findings regarding Malaysia's carbon emissions are statistically valid.

4.2 Stationarity

Augmented Dickey-Fuller Unit Root Test on CO

Null Hypothesis: CO has a unit root
Exogenous: Constant
Lag Length: 0 (Automatic - based on AIC, maxlag=4)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-0.985157	0.7500
Test critical values:	1% level	-3.596616	
	5% level	-2.933158	
	10% level	-2.604867	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(CO)
Method: Least Squares
Date: 12/14/25 Time: 16:55
Sample (adjusted): 1981 2022
Included observations: 42 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
CO(-1)	-0.019533	0.019828	-0.985157	0.3305
C	0.244607	0.113771	2.149992	0.0377

R-squared	0.023689	Mean dependent var	0.140238
Adjusted R-squared	-0.000719	S.D. dependent var	0.268705
S.E. of regression	0.268802	Akaike info criterion	0.256762
Sum squared resid	2.890172	Schwarz criterion	0.339508
Log likelihood	-3.391995	Hannan-Quinn criter.	0.287091
F-statistic	0.970535	Durbin-Watson stat	2.031056
Prob(F-statistic)	0.330469		

Stationary ensures that the statistical properties of a time series such as means and variance do not change over time which is necessary for reliable inference in time series regression. The tool used is EViews, in which the Augmented Dickey-Fuller (ADF) test was conducted for the CO_2 emissions per capita variable.

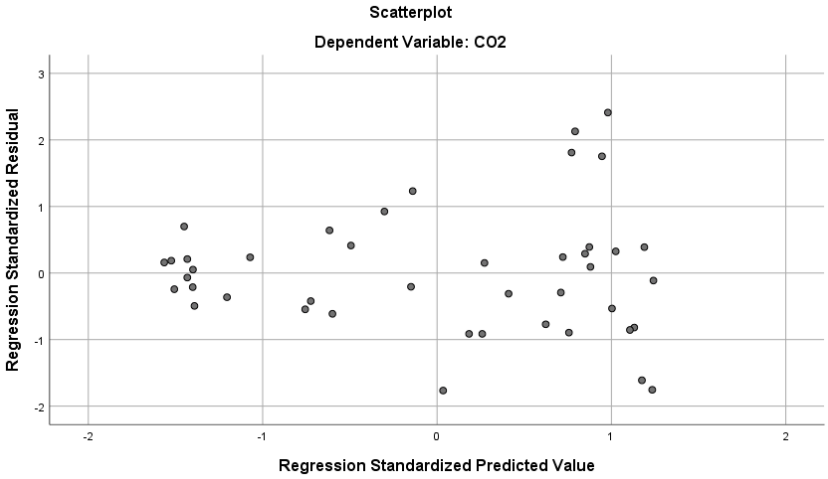
Decision rule:

- p-value < 0.05 = stationary
- p-value ≥ 0.05 = non-stationary

Result:

The ADF test yielded a p-value of 0.7500. Since p-value ≥ 0.05, we fail to reject the null hypothesis of a unit root. This indicates that CO_2 emissions per capita is non-stationary, meaning its mean and are not constant over time.

4.3 Heteroskedasticity



Heteroskedasticity happens when the variance of residuals is not constant across predicted values, thus violating a key OLS assumption.

A scatter plot of unstandardized residual (RES_1) against predicted values (PRE_1) was created in SPSS.

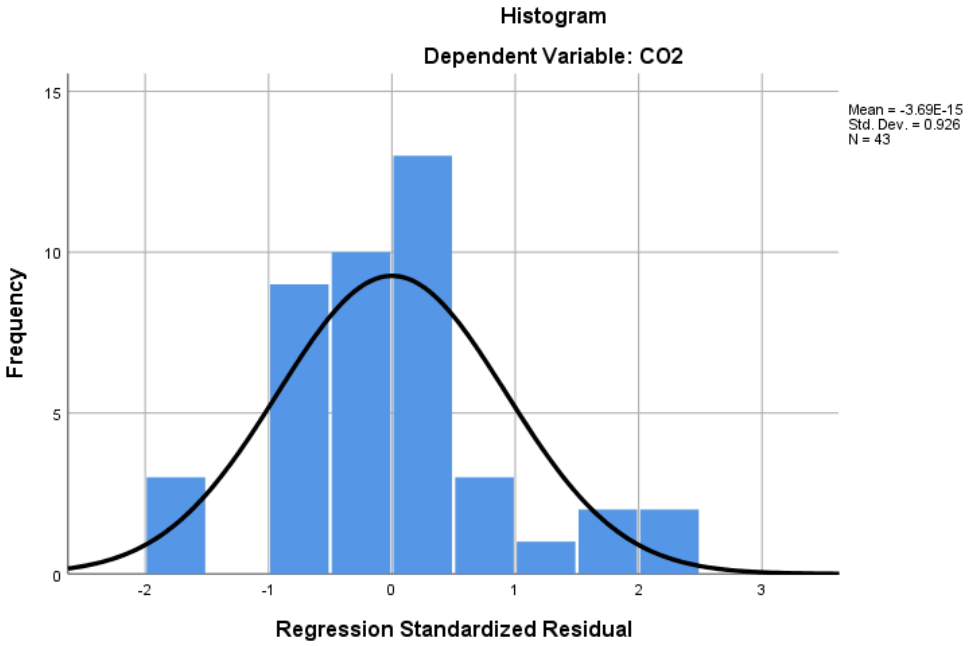
Decision rule:

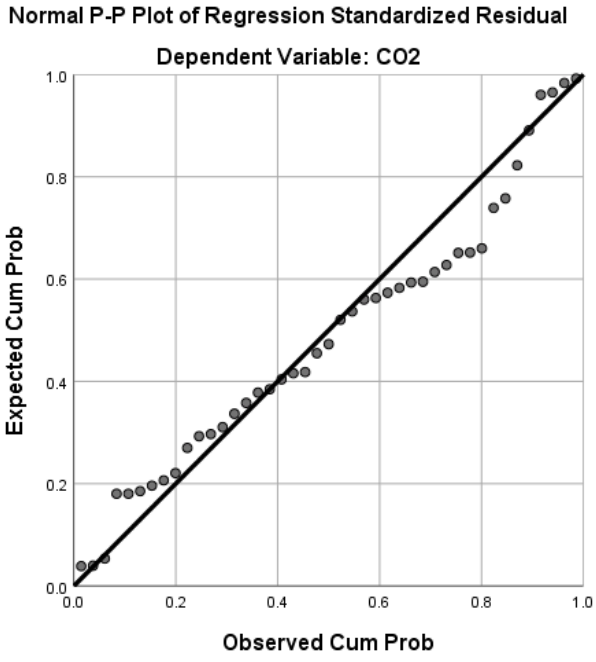
- p-value < 0.05 = heteroskedasticity
- p-value ≥ 0.05 = variance is constant
- Residuals are homoskedastic if the scatter plot shows random dispersion around zero with no clear pattern

Results:

The scatter plots show residuals randomly dispersed around the zero with no shape. Thus, the variance of residuals is constant indicating no heteroskedasticity.

4.4 Normality





Normality ensures that residuals are approximately normally distributed which validates the t-tests and F-test.

A histogram of residual (RES_1) and normal P-P plot were generated in SSPS.

Decision rule:

- p-value ≥ 0.05 = residuals are normal
- p-value < 0.05 = residuals are not normal
- Residuals are considered normally distributed if the histogram is approximately normal bell shaped and the points in the P-P plot line lie close to the diagonal line

Results:

Histogram shows a bell curve and the P-P plot shows residuals close to the diagonal. Therefore, the residuals are approximately normally distributed indicating that the normality assumption is satisfied.

4.5 Multicollinearity

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients			Collinearity Statistics	
		B	Std. Error	Beta	t	Sig.	Tolerance	VIF
1	(Constant)	.291	.957		.304	.763		
	GDP	-5.753E-5	.000	-.067	-.825	.415	.026	38.753
	Energy	.003	.000	.885	11.064	.000	.027	37.507
	Industry	.025	.016	.039	1.601	.118	.290	3.451
	Export	.009	.009	.029	.985	.331	.202	4.959
	Import	-.019	.007	-.101	-2.599	.013	.112	8.919
	Urban	-.231	.108	-.115	-2.128	.040	.058	17.165

a. Dependent Variable: CO2

Multicollinearity arises when independent variables are highly correlated which can inflate standard errors and make coefficient estimates unreliable.

The Variance Inflation Factor (VIF) were obtained from SSPS coefficients table

Decision rule:

- VIF > 10 = multicollinearity exists
- VIF ≤ 10 = acceptable.

Results:

Variance Inflation Factor (VIF) values show that three independent variables exceed 10 indicating that multicollinearity exists while the other three independent variables have VIF below 10 indicating it is acceptable. This indicates that multicollinearity exists in the model and may affect certain coefficients and should be interpreted cautiously.

5.0 REGRESSION ANALYSIS AND DISCUSSION OF FINDINGS

5.1 Constant and Beta Coefficients

The estimated regression equation is :

$CO_2 = 0.291 - 0.00005753(GDP) + 0.003(Energy) + 0.025(Industry) + 0.009(Export) - 0.019(Import) - 0.231(Urban)$

- The constant (0.291) represents the theoretical baseline CO_2 emissions when all independent variables equal zero.

- GDP per capita (−0.00005753): Shows a very small negative effect on CO_2 emissions. A US$1 increase in GDP per capita reduces emissions by 0.00005753 tons, though this is unexpected as economic growth usually increases emissions during industrialization.

- Energy Usage per capita (0.003): Has the largest positive effect. Every 1 kg increase in energy usage raises CO_2 emissions by 0.003 tons, confirming the strong link between fossil fuel consumption and emissions.

- Industry Share (0.025): Positive effect, indicating that a 1% increase in the industrial sector's GDP share increases CO_2 emissions by 0.025 tons per capita, reflecting energy-intensive industrial activities.

- Export Services (0.009): Positive effect, showing that a 1% increase in export transportation services raises emissions by 0.009 tons per capita due to logistics and freight activities.

- Import Services (−0.019): Negative effect, suggesting that a 1% increase in import services reduces emissions by 0.019 tons per capita, possibly due to structural economic changes or changes in imported goods.

- Urbanization (−0.231): Second-largest effect after energy usage. A 1% increase in urbanization growth reduces CO_2 emissions by 0.231 tons per capita, indicating that urban development leads to more efficient infrastructure and resource management.

5.2 Power of Explanation (R^2)

Model Summary[b]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Durbin-Watson
1	.997[a]	.994	.993	.18026	1.688

a. Predictors: (Constant), Urbanization Growth, Industry Shares, Transport Export Services, Transport Import Services, Energy Usage per capita, GDP per capita

b. Dependent Variable: CO2 emissions per capita

- R^2 (Coefficient of Determination) = 0.994:
  The R^2 value indicates that 99.4% of the variation in CO_2 emissions per capita in Malaysia from 1980 to 2022 is explained by the six independent variables. This shows excellent explanatory power, with only 0.6% of variation left unexplained, possibly due to factors like technological changes, environmental policies, climate variations, or measurement errors.

- Adjusted R^2 = 0.993:
  The adjusted R^2 is very close to R^2, differing by only 0.001. This indicates the model is not overfitted, and all variables meaningfully contribute to explaining CO_2 emissions even after adjusting for the number of predictors.

- R (Multiple Correlation) = 0.997:
  The multiple correlation coefficient shows a very strong positive correlation between the independent variables and CO_2 emissions per capita.

- Standard Error of Estimate = 0.18026:
  The standard error represents the average deviation of observed CO_2 values from predicted values. Its small size relative to the dependent variable range (2.11 to 8.0 tons per capita) indicates high prediction accuracy.

5.3 Significance Level for Individual Statistics

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients			Collinearity Statistics	
		B	Std. Error	Beta	t	Sig.	Tolerance	VIF
1	(Constant)	.291	.957		.304	.763		
	GDP per capita	-5.753E-5	.000	-.067	-.825	.415	.026	38.753
	Energy Usage per capita	.003	.000	.885	11.064	.000	.027	37.507
	Industry Shares	.025	.016	.039	1.601	.118	.290	3.451
	Transport Export Services	.009	.009	.029	.985	.331	.202	4.959
	Transport Import Services	-.019	.007	-.101	-2.599	.013	.112	8.919
	Urbanization Growth	-.231	.108	-.115	-2.128	.040	.058	17.165

a. Dependent Variable: CO2 emissions per capita

Three variables are statistically significant at the 5% level ($\alpha = 0.05$):

- Energy Usage per capita ($p < 0.001$):
  Highly significant, with t-statistic = 11.064. We reject the null hypothesis (H_0: $\beta_2 = 0$) and conclude that energy usage has a statistically significant effect on CO_2 emissions.

- Import Services ($p = 0.013$):
  Statistically significant, with t-statistic = -2.599. We reject the null hypothesis (H_0: $\beta_5 = 0$) and conclude that import services have a statistically significant effect on CO_2 emissions.

- Urbanization ($p = 0.040$):
  Statistically significant, with t-statistic = -2.128. We reject the null hypothesis (H_0: $\beta_6 = 0$) and conclude that urbanization has a statistically significant effect on CO_2 emissions.

Four variables are not statistically significant at the 5% level:

- Constant ($p = 0.763$):
  Not significant. We fail to reject H_0.

- GDP per capita ($p = 0.415$):
  Not significant. We fail to reject H_0 (H_0: $\beta_1 = 0$). This may be due to multicollinearity with energy usage (VIF = 38.753).

- Industry Share ($p = 0.118$):
  Not significant. We fail to reject H_0 (H_0: $\beta_3 = 0$). The effect of industry may already be captured through energy consumption.

- Export Services ($p = 0.331$):
  Not significant. We fail to reject H_0 (H_0: $\beta_4 = 0$).

5.4 Significance Level for Overall Statistic

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	189.515	6	31.586	972.029	.000[b]
	Residual	1.170	36	.032		
	Total	190.685	42			

a. Dependent Variable: CO2 emissions per capita

b. Predictors: (Constant), Urbanization Growth, Industry Shares, Transport Export Services, Transport Import Services, Energy Usage per capita, GDP per capita

The F-statistic of 972.029 with a p-value of 0.000 ($p < 0.001$) indicates that the regression model is highly statistically significant. This provides strong evidence to reject the null hypothesis that all coefficients are equal to zero, confirming that the model fits the data substantially better than a model with no predictors.

The R^2 value of 0.994 shows that 99.4% of the variation in CO_2 emissions per capita in Malaysia from 1980 to 2022 is explained by the six independent macroeconomic variables. The residual sum of squares is only 1.170, representing 0.6% of the total variation, while the mean square error (MSE = 0.032) is very small, demonstrating high precision in the model's predictions.

Overall, these results confirm that the regression model is statistically valid and reliable for analyzing the determinants of CO_2 emissions. Energy Usage, Import Services, and Urbanization emerge as the most significant predictors, highlighting the important role of macroeconomic factors in shaping Malaysia's CO_2 emission patterns over the 43-year study period.

6.0 MANAGERIAL IMPLICATIONS AND RECOMMENDATIONS

According to the regression results, Malaysia's CO_2 emissions per capita are statistically significantly influenced by energy consumption per capita, transportation import services, and urbanisation expansion. This suggests that energy use, transit import management, and metropolitan area development have a significant impact on Malaysia's environmental consequences. Malaysia's reliance on fossil fuels continues to be a significant environmental concern, as evidenced by the fact that energy usage per capita has the largest beneficial influence on CO_2 emissions. This emphasises how crucial it is to increase energy efficiency and lessen reliance on carbon-intensive energy sources.

The significant negative relationship between transport import services and CO_2 emissions suggests that changes in import structure or logistics efficiency may help reduce emissions. This may reflect the adoption of more efficient transportation systems or a shift toward importing less carbon-intensive goods.

It is also interesting to observe that the growth in urbanization is inversely related to CO_2 emission, which perhaps underlines a fact that the future Malaysian urban growth could result in improved energy efficiency due to better public transit, infrastructure, and higher population density. This shows how thoughtful urbanization can promote environmental sustainability. On the other hand, there were no statistically significant results for GDP per capita, industrial share, or transportation export services. This indicates that with the proper management of energy use and urban planning, economic growth and industrial activities do not necessarily lead to higher emissions.

Based on the findings, managers and policymakers should prioritize energy efficiency by enforcing clean manufacturing technologies, energy-saving regulations, and renewable energy development to reduce carbon emissions. The government should also promote compact city planning, green buildings, and improved public transport for sustainable urban development. Overall, Malaysia can pursue economic growth alongside stronger environmental regulations, as economic growth is not a significant driver of CO_2 emissions in this model.

7.0 CONCLUSION

Overall, the results demonstrate that energy consumption, transportation import activities, and urbanisation patterns rather than actual economic growth are the primary drivers of CO_2 emissions in Malaysia. The primary cause of emissions is still a large reliance on fossil fuel-based energy, underscoring the significance of increasing energy efficiency and switching to greener energy sources. Environmental pressure can also be lessened by well-planned urban expansion and more effective import logistics. Thus, limiting energy use, enhancing the effectiveness of transportation, and encouraging sustainable urbanisation can successfully lower CO_2 emissions in Malaysia while permitting ongoing economic growth.

8.0 APPENDICES

8.1 Full Dataset (1980–2022)

	GDP/cap (US$)	Energy Usage / capita (kg of oil equivalent)	CO_2 emissions/ capita (ton)	Industry Share (% of GDP)	Transport commercial export services (%)	Transport commercial import services (%)	Urbanization (Annual Growth %)
1980	3216.288	916.0770595	2.11	41.79196	45.73445	45.1273	4.906156
1981	3332.82	945.6881876	2.16	41.05672	50.16465	43.14073	4.951147
1982	3420.217	928.8008109	2.2	38.56669	44.46688	36.23986	4.960612
1983	3518.795	1048.095511	2.39	38.95189	40.24702	37.5834	4.97704
1984	3670.366	1046.185554	2.4	39.2334	39.19918	40.08419	4.988366
1985	3515.255	1019.401324	2.45	39.23426	36.91268	37.62898	4.988311
1986	3444.117	1107.565351	2.36	39.23364	36.84688	44.27434	4.958445
1987	3508.009	1084.810303	2.35	38.52747	42.29116	48.03438	4.878884
1988	3737.26	1102.659972	2.44	38.35661	44.58357	48.37684	4.778607
1989	3953.815	1174.150362	2.78	39.80405	46.58007	40.57393	4.643859
1990	4184.761	1188.82453	3.17	42.19817	46.9294	31.77717	4.531661
1991	4456.931	1433.290442	3.7	42.10651	46.83116	32.42532	4.374127
1992	4726.138	1542.601653	3.79	41.14626	43.8854	32.64157	5.065848
1993	5062.591	1628.856475	4.02	40.08444	36.63613	25.63264	5.027066
1994	5390.102	1579.983076	4.21	40.04124	36.55199	20.34214	4.94898
1995	5771.352	1669.430762	4.43	41.40188	37.84585	21.55624	4.877683
1996	6190.464	1804.84151	4.93	43.52821	31.21346	18.85855	4.79736
1997	6480.556	2053.29505	5.16	44.57354	30.63187	18.37755	4.688775
1998	5857.967	1902.762247	5.05	43.87575	31.51506	19.91998	4.594908
1999	6070.956	1883.08206	5.33	46.45902	32.27919	20.81609	4.457907
2000	6455.614	2102.392125	5.63	48.32057	35.47472	20.13191	4.357598
2001	6334.928	2135.788405	5.79	46.20269	34.67946	19.02255	3.916765
2002	6516.907	2154.461906	6.01	45.11538	36.26088	19.29617	3.892027
2003	6732.92	2254.16676	6.22	46.57866	36.13524	20.46289	3.802879
2004	7025.346	2371.200287	6.59	48.53032	41.19586	18.58497	3.712795
2005	7234.024	2515.148727	6.85	46.36832	38.60051	20.65285	3.628417
2006	7469.811	2472.126833	6.98	46.51566	40.70069	19.80249	3.55318
2007	7769.597	2584.765826	7.47	44.596	38.42733	24.64781	3.464718
2008	7976.118	2678.819498	7.73	45.10507	37.89423	22.02795	3.352735
2009	7700.461	2483.792734	6.86	40.96782	33.99211	15.60426	3.212464
2010	8118.923	2529.591689	7.37	40.49893	31.56657	14.10821	3.057765
2011	8400.322	2591.224708	7.33	39.82427	29.97987	13.06368	2.729391
2012	8710.527	2587.431704	7.29	40.135	26.86993	11.00236	2.628445
2013	8964.807	2869.769116	7.71	39.88537	27.23086	11.1066	2.61164
2014	9341.731	2879.923522	8.02	39.92442	28.22402	11.37178	2.597984
2015	9648.68	2706.926963	7.92	38.44967	26.29935	12.05151	2.593878
2016	9901.48	2759.172283	7.64	37.68327	24.59377	11.71342	2.608628
2017	10293.76	2617.764623	7.26	38.11472	27.07542	12.113	2.572455
2018	10610.2	2839.452448	7.66	38.29144	26.94928	12.66821	2.479396
2019	10902.98	2878.298885	7.61	37.47656	26.45719	12.77953	2.344619
2020	10171.47	2722.603166	7.44	35.93474	29.61316	14.87891	2.052908
2021	10388.27	2793.499535	7.6	37.68269	31.70259	18.76973	1.844773
2022	11174.21	2867.096519	8	39.09483	30.64619	18.84666	1.86226

8.2 Regression Output Tables

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients			Collinearity Statistics	
		B	Std. Error	Beta	t	Sig.	Tolerance	VIF
1	(Constant)	.291	.957		.304	.763		
	GDP per capita	-5.753E-5	.000	-.067	-.825	.415	.026	38.753
	Energy Usage per capita	.003	.000	.885	11.064	.000	.027	37.507
	Industry Shares	.025	.016	.039	1.601	.118	.290	3.451
	Transport Export Services	.009	.009	.029	.985	.331	.202	4.959
	Transport Import Services	-.019	.007	-.101	-2.599	.013	.112	8.919
	Urbanization Growth	-.231	.108	-.115	-2.128	.040	.058	17.165

a. Dependent Variable: CO2 emissions per capita

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	189.515	6	31.586	972.029	.000[b]
	Residual	1.170	36	.032		
	Total	190.685	42			

a. Dependent Variable: CO2 emissions per capita

b. Predictors: (Constant), Urbanization Growth, Industry Shares, Transport Export Services, Transport Import Services, Energy Usage per capita, GDP per capita

Model Summary[b]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Durbin-Watson
1	.997[a]	.994	.993	.18026	1.688

a. Predictors: (Constant), Urbanization Growth, Industry Shares, Transport Export Services, Transport Import Services, Energy Usage per capita, GDP per capita

b. Dependent Variable: CO2 emissions per capita

8.3 Diagnostic Test Results

Model Summary[b]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Durbin-Watson
1	.997[a]	.994	.993	.18026	1.688

a. Predictors: (Constant), Industry, Energy, Export, Import, Urbanization, GDP

b. Dependent Variable: CO2

Augmented Dickey-Fuller Unit Root Test on CO

Null Hypothesis: CO has a unit root
Exogenous: Constant
Lag Length: 0 (Automatic - based on AIC, maxlag=4)

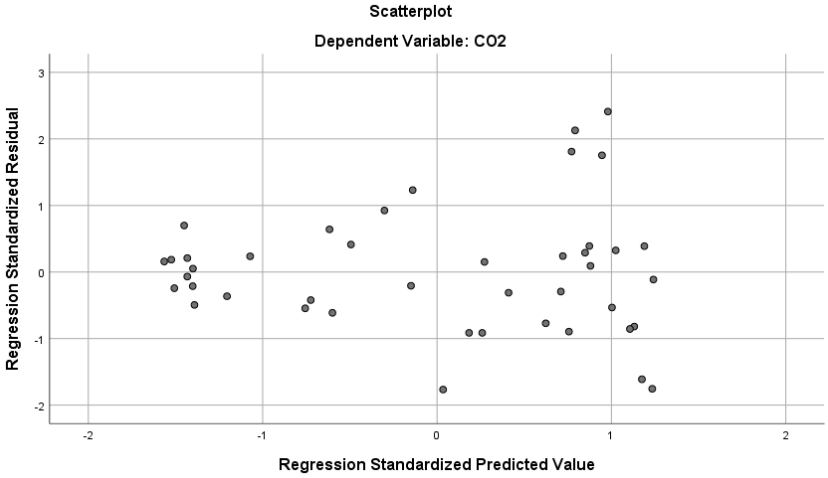
		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-0.985157	0.7500
Test critical values:	1% level	-3.596616	
	5% level	-2.933158	
	10% level	-2.604867	

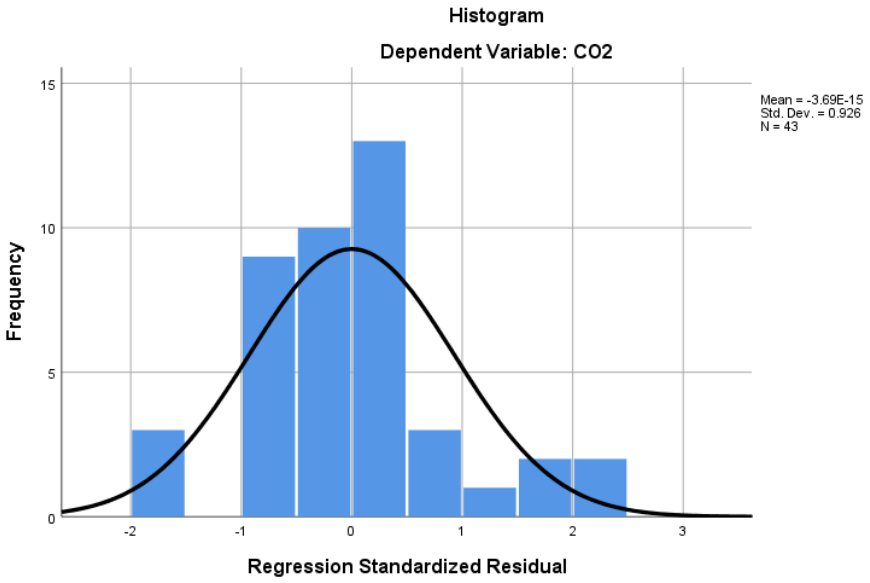
*MacKinnon (1996) one-sided p-values.

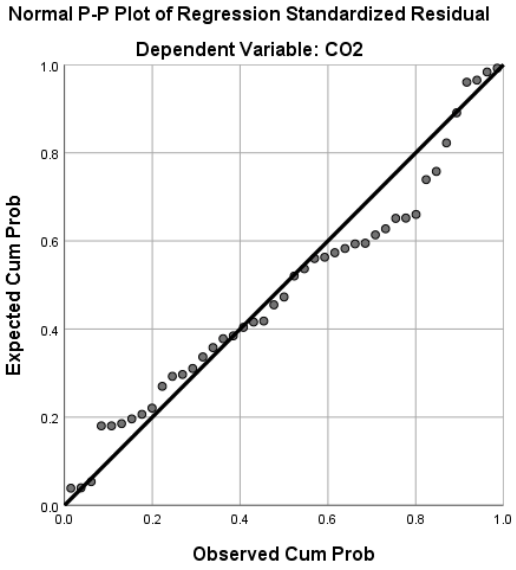
Augmented Dickey-Fuller Test Equation
Dependent Variable: D(CO)
Method: Least Squares
Date: 12/14/25 Time: 16:55
Sample (adjusted): 1981 2022
Included observations: 42 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
CO(-1)	-0.019533	0.019828	-0.985157	0.3305
C	0.244607	0.113771	2.149992	0.0377

R-squared	0.023689	Mean dependent var	0.140238
Adjusted R-squared	-0.000719	S.D. dependent var	0.268705
S.E. of regression	0.268802	Akaike info criterion	0.256762
Sum squared resid	2.890172	Schwarz criterion	0.339508
Log likelihood	-3.391995	Hannan-Quinn criter.	0.287091
F-statistic	0.970535	Durbin-Watson stat	2.031056
Prob(F-statistic)	0.330469		







Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients			Collinearity Statistics	
		B	Std. Error	Beta	t	Sig.	Tolerance	VIF
1	(Constant)	.291	.957		.304	.763		
	GDP	-5.753E-5	.000	-.067	-.825	.415	.026	38.753
	Energy	.003	.000	.885	11.064	.000	.027	37.507
	Industry	.025	.016	.039	1.601	.118	.290	3.451
	Export	.009	.009	.029	.985	.331	.202	4.959
	Import	-.019	.007	-.101	-2.599	.013	.112	8.919
	Urban	-.231	.108	-.115	-2.128	.040	.058	17.165

a. Dependent Variable: CO2

9.0 REFERENCES

Baek, J., Cho, Y., & Koo, W. W. (2009). The environmental consequences of globalization: A country-specific time-series analysis. Ecological Economics, 68(8–9), 2255–2264. Retrieved from https://doi.org/10.1016/j.ecolecon.2009.02.021

Grossman, G. M., & Krueger, A. B. (1995). Economic Growth and the Environment. The Quarterly Journal of Economics, 110(2), 353–377. Retrieved from https://doi.org/10.2307/2118443

Poumanyvong, P., & Kaneko, S. (2010). Does urbanization lead to less energy use and lower CO_2 emissions? A cross-country analysis. Ecological Economics, 70(2), 434–444. Retrieved from https://doi.org/10.1016/j.ecolecon.2010.09.029

Saidi, K., & Hammami, S. (2015). The impact of CO_2 emissions and economic growth on energy consumption in 58 countries. Energy Reports, 1, 62–70. Retrieved from https://doi.org/10.1016/j.egyr.2015.01.003

Stern, D. I. (2004). The Rise and Fall of the Environmental Kuznets Curve. World Development, 32(8), 1419–1439. Retrieved from https://doi.org/10.1016/j.worlddev.2004.03.004

Zhang, X.-P., & Cheng, X.-M. (2009). Energy consumption, carbon emissions, and economic growth in China. Ecological Economics, 68(10), 2706–2712. Retrieved from https://doi.org/10.1016/j.ecolecon.2009.05.011